Reliable Annotation and Evaluation of Malaysian Malay-English Code-Mixed Sentiment Data: A Critical Review of an Unresolved Problem

Ng Yong Vay

[Faculty and University]
E-mail

Article history

Received:
xx xxx 2023

Received in revised form:
xx xxx 2023

Accepted:
xx xxx 2023

Published online:
xx xxx 2023

*Corresponding author
yongvay@gmail.com

Abstract

Malaysian social media users mix English and Malay inside single messages, producing the contact variety known locally as Manglish. The studies reviewed here usually treat sentiment analysis of Malay-English mixed writing as a modelling problem. This review indicates that the more limiting problem originates earlier, in the sentiment labels themselves and in the evidence offered that those labels are sound. The review covers annotation practice and reliability evaluation for Malaysian Malay-English code-mixed sentiment data. Comparative benchmarking of sentiment classifiers is outside its scope. Three contributions follow. First, the annotation strategies used for low-resource code-mixed text are organised into three tiers, and the trade-off between label fidelity and scalability is examined across them. Second, two sources of neutral-label skew are separated, one produced by the labelling tools and the other by genuine uncertainty among annotators. The four Malay-English sentiment resources are then examined against what a reliability claim requires, and none supplies it. Third, the class imbalance in the label distribution is shown to be the condition that makes Cohen's kappa more difficult to interpret. The review closes with a research agenda covering minimum reporting standards, agreement measurement under class imbalance, and the untested assumption that annotators of different types make different errors.

Keywords: *sentiment analysis, code-mixed text, Manglish, annotation reliability, inter-annotator agreement*

1. Introduction

Sentiment analysis, also known as opinion mining, is the automated task of deciding whether a piece of text expresses a positive, negative or neutral opinion (Poria et al., 2023). Organisations use it to measure public opinion at scale, in domains that include consumer products and services, healthcare, political elections and financial services (Nazir et al., 2022). However, the tools used for this task assume that their input is written in one standard language. Joshi et al. (2020) highlight that only a small number of the world's languages are represented in current language technologies, with lexicons, pre-trained models and benchmark corpora concentrated in those few languages. Among these resources, a pre-trained model is one that has already learned general patterns from a large body of text and is then adapted to a particular task such as sentiment analysis (Devlin et al., 2019). Many of these resources are developed primarily for individual languages, so text in which two languages alternate within a single message, a pattern known as code-

* Corresponding author. yongvay@gmail.com

switching, falls outside the conditions for which they were designed. Doğruöz et al. (2021) report that the effect of code-switching on model performance varies by task, with current models performing well on code-switched language identification and part-of-speech tagging but only slightly better than chance on natural language inference. Performance on code-switched text is therefore often lower than performance on monolingual text for the same task, though not uniformly so (Doğruöz et al., 2021; Winata et al., 2023). Malaysian social media text is one setting in which such code-switching is common, and the two languages most often involved are English and Malay (AlAfnan et al., 2025; Treffers-Daller et al., 2022).

Malaysia is a multilingual and multi-ethnic country. Drawing on Ethnologue figures, Coluzzi (2017) records around 140 languages in use in the country, and reports that code-mixing between a speaker's first languages and English or Malay is widespread in everyday communication. On social media, Malaysian users alternate between languages within a single message rather than writing in one language at a time (Treffers-Daller et al., 2022; AlAfnan et al., 2025). English and Malay are among the languages most frequently combined in Malaysian written communication (AlAfnan et al., 2025). Two local terms are used for this code-mixed writing, and they refer to different combinations of languages (AlAfnan et al., 2025; Romadhona et al., 2022). AlAfnan et al. (2025) describe *Manglish* as a hybrid form that combines Malay and English. Romadhona et al. (2022) describe *Bahasa Rojak* as a variety that combines English, Malay and Chinese. This review is concerned with Malay-English mixing. *Manglish* refers to that combination directly, whereas *Bahasa Rojak* refers to a wider combination that also includes Chinese.

Descriptions of *Manglish* in the literature are not uniform. AlAfnan et al. (2025) note that it is sometimes viewed as having a negative effect on standard English and Malay rather than as a legitimate language variety in its own right. Its structure, however, is systematic rather than random. Treffers-Daller et al. (2022) analyse the classroom speech of two Malaysian teachers of English and find that Malay-English switching follows describable syntactic patterns. AlAfnan et al. (2025) report that speakers use *Manglish* for identifiable purposes, which include communicating efficiently, marking identity and building rapport.

The same systematicity appears at the level of spelling. Maskat and Rahman (2020) document the Malay negator *tidak* being written as *x*, and *tidak ada* being written as *takde*, *xde* or *tade*. They sort this variation into five categories and map two of them back to their standard forms, and this categorisation suggests that the variation follows recurring patterns rather than occurring arbitrarily. A dictionary of standard Malay, such as *Kamus Dewan*, does not list these non-standard forms (Dewan Bahasa dan Pustaka, 2005). Tools that rely on such a dictionary may therefore treat them as out-of-vocabulary items (Maskat and Rahman, 2020). Abu Bakar et al. (2020) accordingly describe Malay social media text as misspelled, out-of-vocabulary, ill-formed and noisy. That description reflects the reference standard being applied rather than a lack of care by the writer. These writing conventions are consistent within the observed variation, but they differ from the conventions that standard natural language processing (NLP) tools were designed to handle.

This mismatch between writing conventions and tool design raises the question of where the difficulty in processing such text actually lies. When a sentiment model performs poorly on Malay-English mixed text, one common explanation concerns the model architecture (Doğruöz et al., 2021; Winata et al., 2023). Under this view the model handles code-switched input less effectively than monolingual input, so a larger model, or one further trained on Malay-English or other Southeast Asian data, may perform better. The evidence for this explanation is mixed. Khanuja et al. (2020) report that multilingual models fine-tuned on code-switched data achieve the strongest scores among the systems evaluated on their benchmark. Adilazuarda et al. (2022) find that robustness to code-mixing varies with the embedded language, and attribute the variation to bias in the pre-training corpus. In contrast, Winata et al. (2021) find that pre-trained multilingual models do not necessarily guarantee high-quality representations of code-switched text. Conneau et al. (2020) describe a trade-off between positive transfer and capacity dilution. Capacity dilution refers to the reduction in the modelling capacity available to each individual language as more languages are added to a model of fixed size. A model required to accommodate many languages may therefore produce representations of lower quality for each one. Doğruöz et al. (2021) identify a further limitation that is not architectural, in that the multilingual models in common use are trained mainly on monolingual corpora and may receive little exposure to code-switched text during pre-training. Model choice therefore accounts for part of the difficulty but not all of it.

Another source of difficulty arises one step earlier, at the point where sentiment labels are created. Supervised sentiment analysis requires a sentiment label for every message before a model can be trained or evaluated on it. Poria et al. (2023) note that this heavy reliance on annotated data is a limitation, and that it has renewed interest in supplying models with external knowledge such as sentiment lexicons. That label is assigned either by a human annotator or by an automatic labelling method, and this step is called annotation. The labels form the supervision signal used to train a model, and the labels in a test set are the standard against which the model's predictions are compared. Incorrect labels can affect both model training and subsequent evaluation. During training, a model may learn the patterns associated with incorrect labels, which can reduce predictive accuracy (Frénay and Verleysen, 2014). During evaluation, incorrect labels in a test set can bias the measurement of model performance, and Northcutt et al. (2021) show that label errors can change which of two models a benchmark ranks higher. Bowman and Dahl (2021) name annotation reliability as one of four conditions a sound evaluation benchmark must meet. An incorrect label can therefore have an effect at two separate stages, first when the model is trained and again when its performance is measured.

Three findings, each developed in the sections that follow, point to the annotation label rather than the model as the more limiting constraint for Malaysian Malay-English code-mixed sentiment data at present. They also indicate that the evidence offered for the soundness of those labels is limited. First, this review identifies four Malay-English sentiment resources (Kasmuri and Basiron, 2019; Romadhona et al., 2022; Kong et al., 2023; Mohd Shamsuddin et al., 2024). None of them is at the same time large, meaning at least ten thousand labelled messages, balanced across

sentiment classes, drawn from casual social media, and accompanied by an assessment of annotation reliability by human annotators. Each of them satisfies some of these criteria but not all of them. Second, the class imbalance in the largest of them appears to originate in the labelling method rather than in Malaysian discourse (Mohd Shamsuddin et al., 2024; Kong et al., 2023). Third, annotation reliability is rarely reported across these four resources (Kasmuri and Basiron, 2019; Romadhona et al., 2022; Kong et al., 2023; Mohd Shamsuddin et al., 2024). Where an inter-annotator agreement coefficient is reported, it was measured on a two-way judgement rather than on the three-way positive, negative and neutral labelling for which the resource is used.

The second and third findings are closely related rather than independent. Reliability is normally assessed by having independent annotators label the same text and computing a chance-corrected agreement coefficient, such as Cohen's kappa (κ) (Cohen, 1960; Artstein and Poesio, 2008). Cohen's κ becomes difficult to interpret when one category dominates the label distribution, because the agreement expected by chance rises with that skew (Feinstein and Cicchetti, 1990; Byrt et al., 1993). Di Eugenio and Glass (2004) call this the prevalence problem. The skew that the labelling methods produce is therefore also the condition that can make the statistic used to assess the labels more difficult to interpret. Correcting either weakness alone leaves the other in place. This shortage of reliably labelled data is not confined to the four resources examined here. Selvaraju et al. (2024) report that comprehensive datasets and annotated resources for code-mixed Malay text are still lacking.

These three findings lead to the question this review examines: why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data have not yet been established. Linguistic description, model architecture and application domain are discussed only where they are relevant to that question. Two related topics fall outside the scope. The first is comparative benchmarking of sentiment classifiers, and the second is sentiment analysis as a general field. Both topics have already been surveyed in detail (Poria et al., 2023; Nazir et al., 2022; Selvaraju et al., 2024). This review does not propose, implement or evaluate a system, and it does not report the authors' own experimental work. Its contribution is an analysis of why reliable annotation and evaluation remain difficult to establish for this data, together with a set of directions for future research.

This review makes three contributions:

1. It organises the annotation strategies used for low-resource code-mixed text into three tiers, which are manual, semi-automated and fully automated labelling, and examines the trade-off between label fidelity and scalability across them.
2. It distinguishes two sources of neutral-label skew, one produced by the labelling tools themselves and the other by genuine uncertainty among annotators. It also examines what the studies introducing the four Malay-English sentiment resources report about the reliability of their annotation.

3. It shows how the linguistic, annotation and evaluation challenges compound one another rather than merely co-occurring, and sets out a research agenda that follows from that interaction.

References

Abu Bakar, M. F. R., Idris, N., Shuib, L., and Khamis, N. (2020). Sentiment analysis of noisy Malay text: State of art, challenges and future work. IEEE Access, 8, 24687–24696. https://doi.org/10.1109/ACCESS.2020.2968955

Adilazuarda, M. F., Cahyawijaya, S., Winata, G. I., Fung, P., and Purwarianti, A. (2022). IndoRobusta: Towards robustness against diverse code-mixed Indonesian local languages. Proceedings of the First Workshop on Scaling Up Multilingual Evaluation (SUMEval), 25–34.

AlAfnan, M. A., MohdZuki, S. F., and AlAfnan, S. M. (2025). Blending identities: Exploring the linguistic, cultural, and pragmatic dimensions of Manglish in the Malaysian society. Studies in Media and Communication, 13(3). https://doi.org/10.11114/smc.v13i3.7654

Artstein, R., and Poesio, M. (2008). Survey article: Inter-coder agreement for computational linguistics. Computational Linguistics, 34(4), 555–596. https://doi.org/10.1162/coli.07-034-R2

Bowman, S. R., and Dahl, G. (2021). What will it take to fix benchmarking in natural language understanding? Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 4843–4855. https://doi.org/10.18653/v1/2021.naacl-main.385

Byrt, T., Bishop, J., and Carlin, J. B. (1993). Bias, prevalence and kappa. Journal of Clinical Epidemiology, 46(5), 423–429. https://doi.org/10.1016/0895-4356(93)90018-V

Cohen, J. (1960). A coefficient of agreement for nominal scales. Educational and Psychological Measurement, 20(1), 37–46. https://doi.org/10.1177/001316446002000104

Coluzzi, P. (2017). Language planning for Malay in Malaysia: A case of failure or success? International Journal of the Sociology of Language, 2017(244), 17–38. https://doi.org/10.1515/ijsl-2016-0055

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., and Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 8440–8451. https://doi.org/10.18653/v1/2020.acl-main.747

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), 4171–4186. https://doi.org/10.18653/v1/N19-1423

Dewan Bahasa dan Pustaka. (2005). *Kamus Dewan* (4th ed.). Kuala Lumpur: Dewan Bahasa dan Pustaka.

Di Eugenio, B., and Glass, M. (2004). The kappa statistic: A second look. Computational Linguistics, 30(1), 95–101. https://doi.org/10.1162/089120104773633402

Doğruöz, A. S., Sitaram, S., Bullock, B. E., and Toribio, A. J. (2021). A survey of code-switching: Linguistic and social perspectives for language technologies. Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), 1654–1666. https://doi.org/10.18653/v1/2021.acl-long.131

Feinstein, A. R., and Cicchetti, D. V. (1990). High agreement but low kappa: I. The problems of two paradoxes. Journal of Clinical Epidemiology, 43(6), 543–549. https://doi.org/10.1016/0895-4356(90)90158-L

Frénay, B., and Verleysen, M. (2014). Classification in the presence of label noise: A survey. IEEE Transactions on Neural Networks and Learning Systems, 25(5), 845–869. https://doi.org/10.1109/TNNLS.2013.2292894

Joshi, P., Santy, S., Budhiraja, A., Bali, K., and Choudhury, M. (2020). The state and fate of linguistic diversity and inclusion in the NLP world. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 6282–6293. https://doi.org/10.18653/v1/2020.acl-main.560

Kasmuri, E., and Basiron, H. (2019). Building a Malay-English code-switching subjectivity corpus for sentiment analysis. International Journal of Advanced Soft Computing and Its Applications, 11(1), 112–130.

Khanuja, S., Dandapat, S., Srinivasan, A., Sitaram, S., and Choudhury, M. (2020). GLUECoS: An evaluation benchmark for code-switched NLP. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 3575–3585. https://doi.org/10.18653/v1/2020.acl-main.329

Kong, J. T. H., Juwono, F. H., Ngu, I. Y., Nugraha, I. G. D., Maraden, Y., and Wong, W. K. (2023). A mixed Malay-English language COVID-19 Twitter dataset: A sentiment analysis. Big Data and Cognitive Computing, 7(2), 61. https://doi.org/10.3390/bdcc7020061

Maskat, R., and Rahman, N. A. (2020). Categorization of Malay social media text and normalization of spelling variations and vowel-less words. International Journal on Advanced Science, Engineering and Information Technology, 10(4), 1380–1386.

Mohd Shamsuddin, A. M., Samson Juan, S., Chua, S., and Bramantoro, A. (2024). Semi-automatic sentiment identification for Malay-English code-switched data. Journal of Advanced Research Design, 123(1), 198–212. https://doi.org/10.37934/ard.123.1.198212

Nazir, A., Rao, Y., Wu, L., and Sun, L. (2022). Issues and challenges of aspect-based sentiment analysis: A comprehensive survey. IEEE Transactions on Affective Computing, 13(2), 845–863. https://doi.org/10.1109/TAFFC.2020.2970399

Northcutt, C. G., Athalye, A., and Mueller, J. (2021). Pervasive label errors in test sets destabilize machine learning benchmarks. Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS): Datasets and Benchmarks Track.

Poria, S., Hazarika, D., Majumder, N., and Mihalcea, R. (2023). Beneath the tip of the iceberg: Current challenges and new directions in sentiment analysis research. IEEE Transactions on Affective Computing, 14(1), 108–132. https://doi.org/10.1109/TAFFC.2020.3038167

Romadhona, N. P., Lu, S.-E., Lu, B.-H., and Tsai, R. T.-H. (2022). BRCC and SentiBahasaRojak: The first Bahasa Rojak corpus for pretraining and sentiment analysis dataset. Proceedings of the 29th International Conference on Computational Linguistics (COLING), 4418–4428.

Selvaraju, S., Sjarif, N. N. A., and Anuar, M. S. M. (2024). Sentiment analysis in code mixing Malay text: A review. 2024 5th International Conference on Artificial Intelligence and Data Sciences (AiDAS), 280–285. https://doi.org/10.1109/AiDAS63860.2024.10730298

Treffers-Daller, J., Majid, S., Thai, Y. N., and Flynn, N. (2022). Explaining the diversity in Malay-English code-switching patterns: The contribution of typological similarity and bilingual optimization strategies. Languages, 7(4), 299. https://doi.org/10.3390/languages7040299

Winata, G. I., Aji, A. F., Yong, Z. X., and Solorio, T. (2023). The decades progress on code-switching research in NLP: A systematic survey on trends and challenges. Findings of the Association for Computational Linguistics: ACL 2023, 2936–2978. https://doi.org/10.18653/v1/2023.findings-acl.185

Winata, G. I., Cahyawijaya, S., Liu, Z., Lin, Z., Madotto, A., and Fung, P. (2021). Are multilingual models effective in code-switching? Proceedings of the Fifth Workshop on Computational Approaches to Linguistic Code-Switching, 142–153. https://doi.org/10.18653/v1/2021.calcs-1.20